Acute Apnea Does Not Improve 3-km Cycling Time Trial Performance

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ABSTRACT

BOUTEN, J., A. L. COLOSIO, G. BOURGOIS, L. LOOTENS, P. VAN EENOO, J. G. BOURGOIS, and J. BOONE. Acute Apnea Does Not Improve 3-km Cycling Time Trial Performance. *Med. Sci. Sports Exerc.*, Vol. 52, No. 5, pp. 1116–1125, 2020. **Purpose:** Intense exercise evokes a spleen contraction releasing red blood cells into blood circulation. The same mechanism is found after acute apnea, increasing hemoglobin concentration ([Hb]) by 2% to 5%. The aim of this study was twofold: [1] to identify the optimal apnea modalities to acutely increase [Hb] and [2] use these modalities to examine whether prerace apnea can improve a 3-km time trial (TT). **Methods:** In part 1, 11 male subjects performed 12 different apnea protocols based on three modalities: mode, frequency, and intensity. Venous blood samples for [Hb] were collected before, immediately, and 5 min after each protocol. In part 2, 12 recreationally active subjects performed 3-km cycling TT in three different conditions: apnea, control, and placebo, after a 10-min warm-up. Power output, HR, and oxygen uptake ($\dot{V}O_2$) were continuously measured. Venous [Hb] was sampled at baseline, after warm-up, and before TT. Additionally, these subjects performed constant cycling at Δ25 (25% between gas exchange threshold and $\dot{V}O_2$ max) in two conditions (control and apnea) to determine $\dot{V}O_2$ kinetics. **Results:** Although including one single apnea in the warming up evoked a positive change in [Hb] pattern ($P = 0.049$) and one single apnea seemed to improve $\dot{V}O_2$ kinetics in constant submaximal cycling (τ: $P = 0.060$, mean response time: $P = 0.064$), performance during the 3-km TT did not differ between conditions ($P = 0.840$; apnea, 264.8 ± 14.1 s; control, 263.9 ± 12.9 s, placebo, 264.0 ± 15.8 s). Average normalized power output ($P = 0.584$) and $\dot{V}O_2$, HR, and lactate did not differ either ($P > 0.05$). **Conclusions:** These results suggest that potential effects of apnea, that is, speeding of $\dot{V}O_2$ kinetics through a transient increase in [Hb], are overruled by a warming-up protocol. **Key Words:** BREATH HOLDING, SPLEEN CONTRACTION, HEMOGLOBIN, OXYGEN UPTAKE KINETICS, PERFORMANCE

An efficient transport and supply of oxygen (O_2) to organs and muscles is an important physiological determinant of sports performance. Elite athletes performing in disciplines with a predominant aerobic energy turnover continuously try to optimize the oxygen delivery capacity to the muscle. This is largely dependent on the oxygen carrying capacity of the blood (1) of which hemoglobin (Hb) is an important parameter. Indeed, increasing total Hb mass by 1 g improves maximal oxygen consumption ($\dot{V}O_2$ max) by approximately 4 mL·min^{-1} (2). Given that Hb concentration ([Hb]) can increase by up to 13 g·L^{-1} individually (3) and assuming a blood volume of 5 L, this would lead to an absolute increase in maximal oxygen uptake of up to 260 mL·min^{-1}. Acute manipulations, such as blood transfusion and phlebotomy, both increasing and lowering Hb content, have been shown to change $\dot{V}O_2$ max as well as exercise performance (4,5).

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In this context, apnea has been proposed as a potential new method to improve both immediate- and long-term sport performance (6). Acute apnea is known to evoke a spleen contraction releasing red blood cells (RBC) stored in the spleen into blood circulation. As such, [Hb] and hematocrit [Hct] immediately increase by 2% to 5% (3,7,8) and return to baseline 10 min after apnea (9). Similar responses are seen during intense exercise (10) with stronger contractions at higher exercise intensities (11). Barcroft and Florey and Barcroft and Stephens (12,13) were the first to acknowledge the importance of the spleen to performance ability in both dogs and horses. The question, therefore, arises whether acute apnea can be beneficial to improve immediate sports performance in humans. The rationale for this hypothesis is that, by inducing a spleen contraction before exercise, subjects start performance with an advantage in Hb and thus oxygen transport capacity. Increasing the O_2 supply (e.g., by hyperoxia or erythropoietin [EPO]) to the exercising muscles has already shown to accelerate oxygen uptake ($\dot{V}O_2$) kinetics, especially above gas exchange threshold (GET) (14,15) and might thus improve performance (1,5).

Currently, only two studies have investigated the link between acute apnea and performance. Sperlich et al. (16) were the first to examine the effect of acute apnea on a 4-km time trial (TT) performance. They did not observe differences between the apnea and control condition. Moreover, although not statistically significant, it took the subjects 16 extra seconds to complete the TT in the apnea condition. However, when looking at the split times, subjects already lost 7 s in the first kilometer, possibly indicating that they were not ready to perform yet. Indeed, it was reported that subjects started the TT 30 to 45 s after the last apnea, a time interval which could be too short to recover from a physically and mentally demanding maximal breath hold. This problem was partially resolved in the study of Robertson et al. (17) in which subjects started a 400-m freestyle swim 2 min after a series of three apneas. In comparison with Sperlich et al. (16), Robertson et al. (17) also added a warming-up to the design, which corresponds better with the real-life sport context. Yet, no significant differences in completion time were observed between apnea and control. However, no differences in [Hb] were observed either, raising the question whether the appropriate apnea modalities were applied. In the field of apnea, scientists have applied various apnea modalities to induce spleen contraction. The arguments for protocol selections have, however, not been well documented and guidelines to design apnea protocols are thus far lacking.

The aim of this study was, therefore, twofold. In the first part, we aimed to determine the apnea modalities with the highest potential to improve exercise performance: a strong increase in [Hb] combined with a low negative impact on physical and perceived readiness. In the second part, these modalities were added to a warming up to evaluate the value of acute apnea for sport performance by examining whether the physiological changes after apnea lead to better performance in a 3-km TT and improved underlying physiological mechanisms.

METHODS

Ethical Approval

All protocols and procedures were in conformity to the Declaration of Helsinki and approved by the ethical committee of the Ghent University Hospital (EC UZG 2018/0221). Each participant was informed about the procedure and the aim of the study and gave their written informed consent. Both a medical history questionnaire and medical clinical examination were completed before the study. All volunteers were declared to be in good health.

Experimental Design

The experimental protocol for this study consisted of two parts. The rationale for both parts was to make the design transferrable to apply in a competitive setting. In the first part, the aim was to identify the modalities of acute apnea with the strongest increase in [Hb] without affecting readiness to perform exhaustive exercise. We investigated the effect of acute apnea on [Hb] combining three different modalities, that is, mode (static or dynamic), frequency (1 apnea, a series of 5 apneas or a series of 5 followed by 1 separate apnea), and intensity (maximal or 80% of maximal breath hold time). By choosing both heavy protocols eliciting maximal spleen responses and lighter protocols which are better from a practical point of view, we wanted to make it possible to select an optimal protocol to apply in a competitive setting. This means balancing a (near) maximal physiological response with both the practical applicability and low possible negative physiological and psychological effects of acute apnea on performance. In part 2, the effect of prior acute apnea on performance, a 3-km TT, and the underpinning physiological mechanisms (e.g., $\dot{V}O_2$ kinetics) were examined. All tests were performed at the Sport Science Laboratory Jacques Rogge (Ghent University, sea level) with a constant ambient air temperature of 18°C and humidity of 45%.

Subjects

Eleven healthy, male subjects participated in the first part and 12 subjects in part 2. Three subjects participated in both parts. Inclusion criteria for selection were: age (18–30 yr), sex (male), physically active and good general health. Exclusion criteria were: use of nicotine, serving as blood donor within 3 months before the study, experience with breath hold–related sports (i.e., synchronized swimming, breath hold diving, water polo etc.) and residing at altitude in the 3 months before the study. All participants were physically active and performed recreational physical exercise on a self-reported basis of 6.2 ± 3.8 $h \cdot wk^{-1}$ for part 1 and 6.4 ± 3.9 $h \cdot wk^{-1}$ for part 2. Age, anthropometrics, and physiological characteristics are expressed in Table 1. Subjects were instructed not to serve as blood donor nor participate in other studies and to maintain the same level of physical exercise during the intervention period as before the study.

Procedures and Measurements

Part 1: In search of the optimal apnea modalities. ***General procedure.*** The participants visited the laboratory on 14 separate occasions. Tests were scheduled on the same time of the day (±30 min) for each subject to minimize variability in blood values due to the circadian rhythm. All tests took place between 08.00 AM and 12.00 noon and were separated by 48 h. Before each visit to the laboratory, participants were instructed to avoid heavy exercise and to maintain a similar diet for 24 h. During the first visit, all subjects underwent a medical examination and ramp incremental exercise (30 $W \cdot min^{-1}$, starting from a 50-W baseline) to exhaustion to determine peak

TABLE 1. Age, anthropometric records, and results of ramp incremental exercise test and apnea test for part 1 (*n* = 11) and part 2 (*n* = 12).

	Part 1		Part 2	
	Mean	SD	Mean	SD
Age (yr)	22.1	1.5	21.6	1.2
Height (cm)	180.1	4.9	178.5	8.4
Weight (kg)	77.6	7.0	72.1	6.2
HR peak (bpm)	193	7	195	8
$\dot{V}O_2$ peak ($mL \cdot min^{-1} \cdot kg^{-1}$)	55.3	4.1	52.5	5.1
$\dot{V}O_2$ peak ($mL \cdot min^{-1}$)	4254	263	3778	399
P at GET (W)			159	35
P at RCP (W)			263	42
PPO (W)	394	22	373	45
P at Δ25 (W)			207	34
Static apnea time (s)	154	36	131	41

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power output (PPO) and peak oxygen uptake ($\dot{V}O_{2peak}$, Jaeger, Oxycon Pro; Viasys Healthcare GmbH, Höchberg, Germany). During the second visit, subjects performed familiarization tests to get used to performing apneas. The subsequent 12 visits consisted of 12 different apnea conditions, built around three modalities (Mode, Frequency and Intensity, Fig. 1A) and were performed in a randomized order. All cycling tests were performed on a Cyclus 2 Ergometer (RBM Elektronik-Automation, Leipzig, Germany).

Familiarization test. A familiarization apnea tests was included during the second visit. Subjects performed two series, one static and one dynamic, of five maximal apneas interspersed by 2-min rest intervals. The static familiarization was completed in an upright sitting position, whereas during the dynamic familiarization, subjects were cycling at 25% of their PPO as determined from the maximal incremental exercise. The two series were performed in a randomized order with a rest interval of 30 min. Subjects were notified 30 s before each apnea and started the attempt after a 10-s count down. All apneas were preceded by a deep, but not maximal inspiration. During the breath holds, participants were motivated with verbal time cues and strong verbal encouragement. No hyperventilation before the breath holding was allowed to avoid hypoxic syncope. Apneic times were recorded using a chronometer.

Apnea tests. The 12 apnea trials consisted of six static conditions in which the subjects in a seated position, and six dynamic conditions in which subjects cycled at 25% PPO. These were performed on 12 separate days. The order was randomized to avoid possible training and order effects. The same instructions for breath holding were used as in the familiarization tests. Each apnea condition consisted of three modalities: mode (static, dynamic), frequency (one repetition, a series of five or a series of five followed by 1 single breath hold), and intensity (80% of maximal breath hold duration from familiarization or maximal duration). A visual overview of the constitution of the 12 protocols is presented in Figure 1A. The condition in which an extra apnea was performed (5 + 1×), had a 10-min rest interval between the series of fifth and sixth apnea. These modalities consisted of a series of five maximal static or dynamic apneas followed by one static apnea at either 80% or maximal intensity. This procedure could fit well within a warming-up protocol before performance.

Venous blood samples (K3 EDTA 4 mL, Vacutest Kima, Arzergrande PD, Italy) were collected from the V. Mediana Cubiti at baseline, immediately after and 5 min after the last breath-hold attempt. Subjects sat down for at least 15 min before the baseline blood sample to minimize the effects of body posture on the blood values. In the dynamic apnea trials, a second baseline sample was collected after 10 min of cycling at 25% PPO to expose the effect of exercise on hematological parameters. Blood samples were stored cool during storage and transport (2°C–12°C) and analyzed within 48 h postsampling. The samples were sent to a WADA-accredited laboratory (Doping Control Laboratory, Ghent) for analysis of [Hb] and Hct, using an automated hematology analyzer (SYSMEX XT2000i; Sysmex Corporations, Kobe, Japan).

Subjects were also asked to score their level of readiness to compete in a competition on the perceived readiness rating scale from zero to five (18) immediately after as well as 5 min after the last apnea. For evaluating the subjective load of each apnea modality, the Borg (6 to 20) scale was used (19).

During all tests, subjects were closely monitored for cerebral oxygenation using the NIRO 200NX (Hammamatsu Phototonics, Hamamatsu, Japan). If absolute cerebral oxygen saturation fell below 50% or decreased by 20%, subjects were instructed to resume breathing to avoid hypoxic syncope.

Part 2: The influence of apnea on performance. ***General procedure.*** Subjects visited the laboratory on 10 different occasions spread over 5 wk (Fig. 1B). The chosen apnea protocol to evaluate the effect on performance was one maximal static apnea as this turned out to be the optimal protocol based on the results in part 1. During the first visit, all subjects underwent a medical examination and ramp incremental exercise (30 W·min^{-1}) to exhaustion to determine their PPO and peak oxygen uptake ($\dot{V}O_{2peak}$). On the second and third visits, subjects performed two familiarization tests of the 3-km

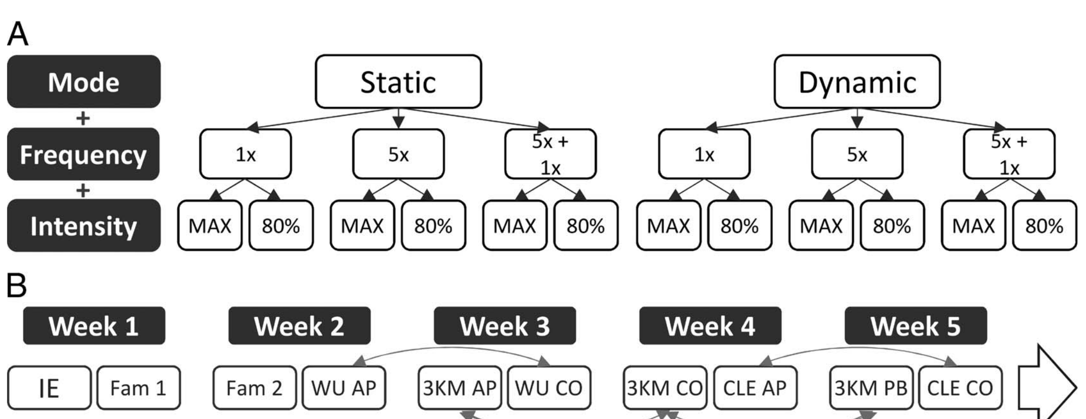

FIGURE 1—Overview of the protocols for part 1 (panel A) and part 2 (panel B). Panel A shows the 12 different apnea protocols built around three modalities: Mode, Frequency and Intensity (Part 1) which were performed in a randomized order. Panel B represents an overview of the 10 test moments in Part 2. IE, incremental exercise; Fam, familiarization; WU, warming up; 3KM, 3 km TT. *Arrows* indicate randomization of the conditions (AP, apnea; CO, control; PB, placebo).

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TT. Additionally, three 3-km TT were performed on different test days (visit 5, 7, and 9) in three different conditions (apnea, placebo, and control) to determine the impact of apnea on performance. Two constant load sessions were performed on two additional test days (visits 8 and 10, apnea and control) to determine $\dot{V}O_2$ and muscle deoxyhemoglobin (HHb) kinetics. Tests were performed in a randomized order. The warming-up was repeated on separate test days to prevent blood sampling before the actual 3-km TT and was meant to evaluate effect of apnea on [Hb] in the rest period after warming up. Tests that were repeated in different conditions, were performed at the same time of the day and same day of the week to avoid influences of diurnal variation and individual weekly schedules. Tests days were separated by at least 48 h. A visual overview of the general procedure is shown in Figure 1B. All cycling tests were performed on a Cyclus 2 Ergometer (RBM Elektronic-Automatik).

Ramp incremental exercise. After a 3-min baseline at 50 W, work load increased linearly with 30 $W \cdot min^{-1}$. Subjects were instructed to keep cadence between 80 and 100 rpm. The test was terminated at volitional exhaustion, that is, when subjects could not keep their cadence above 80 rpm for more than 5 consecutive seconds. Pulmonary gas exchange ($\dot{V}O_2$, $\dot{V}CO_2$) was measured on a breath-by-breath basis (Jaeger Oxycon Pro, Viasys Healthcare GmbH).

Familiarization tests. On the first day of familiarization, subjects started with a series of 5 maximal apneas with 2-min rest intervals to determine maximal apneic duration. Afterwards, subjects performed a 3-km TT at a self-selected gear and pacing strategy. On the second day of familiarization, subjects only performed a 3-km TT.

3 km TT. Time trials were repeated in three different conditions in a randomized order: control, placebo and apnea. Subjects started with 10 min of warming up at the GET interspersed with three 1-min bouts at the respiratory compensation point (RCP), followed by a 10-min resting period before starting the 3-km TT. The difference between the conditions was situated in the 10-min resting period. In the control condition, subjects sat down the entire 10 min. In the placebo condition, subjects were wearing a facemask and were told that they were inspiring a hyperoxic gas mixture for 2 min with the aim to improve sports performance, whereas subjects actually inspired ambient air. This condition was implemented to examine possible placebo effects (20). During the apnea trial, subjects performed one single maximal apnea starting between 5 and 6 min after the warming up and started the TT exactly 3 min after apnea. Starting times for apnea were estimated based on familiarization and warming up times to make sure that subjects started the TT 10 min (±30 s) after warming up. HR (H7 Sensor, Polar, Kempele, Finland) and pulmonary gas exchange (Jaeger Oxycon Pro; Viasys Healthcare GmbH) were continuously measured, with baseline measures starting 3 min before TT. Lactate was measured at the fingertip 1 min before and 1 min after the TT (Lactate Scout; RBM Elektronic-Automatik). Time, continuous, and average power and cadence were recorded by the Cyclus 2 Ergometer Software. To avoid blood collection disturbing the preparation of the 3-km TT, the warming up was performed separately on two additional test days in the apnea and control condition. Venous blood samples (K3 EDTA 4 mL Vacutest Kima; Arzergrande) were taken from the V. Mediana Cubiti at baseline as well as immediately and 10 min post–warming up. The baseline sample was drawn after 15 min of seated rest to minimize the effect of plasma volume changes on [Hb],

Constant load exercises. In addition to the TT, eight subjects performed constant load cycling on two different days in two conditions (apnea and control) in a randomized order. Subjects performed a 3-min baseline at 50 W followed by 6 min at a power output of Δ25 (the power output that corresponds to the value at 25% of the interval between GET and $\dot{V}O_2$ peak) which was on average 207 ± 34 W in this population. In the apnea condition, the 3-min baseline cycling at 50 W started immediately after a single maximal static apnea. Both conditions were repeated 3 times interspersed with 15-min rest intervals.

Subjects were allowed to cycle at a self-selected cadence but were instructed to maintain the cadence throughout all trials. HR (H7 Sensor; Polar), pulmonary gas exchange (Jaeger Oxycon Pro; Viasys Healthcare GmbH) and peripheral tissue oxygenation using near-infrared spectroscopy (NIRS) technology (Oxiplex TS, ISS, Champaign, IL) were continuously measured. For NIRS measurements, the probe was placed longitudinally at the distal section of the muscle belly of the right Musculus Vastus Lateralis, after shaving and cleaning.

Data Analysis

Acute Hb response. The amplitude of the [Hb] change was expressed as Δ0 and Δ5. Δ values for static conditions were calculated by subtracting baseline values for [Hb] immediately before apnea from the value immediately after (Δ0) and 5 min (Δ5) after apnea. For dynamic conditions, the baseline value after 10 min of cycling (BL_{dyn}) was subtracted from the value immediately (Δ0) and 5 min (Δ5) postapnea to correct for the plasma volume change through cycling. Only for the dynamic 5 + 1 conditions (D5 + 1 and D5 + 80), the value was subtracted from the seated baseline as the subjects were already sitting for 12 to 15 min and thus blood plasma volumes already returned toward baseline.

Ramp incremental exercise. Peak power output was determined as the highest value obtained during the ramp test, whereas $\dot{V}O_2$ peak was defined as the highest 30 s average observed during the ramp test.

GET and RCP were derived from the maximal incremental test by three independent researchers. Gas exchange threshold was defined as the point where 1) $\dot{V}CO_2$ increases disproportionate to $\dot{V}O_2$, 2) the first deviation of the linear increase in minute ventilation ($\dot{V}_E$) occurs and 3) an increase in $\dot{V}_E/\dot{V}O_2$ without simultaneous increase in $\dot{V}_E/\dot{V}CO_2$ is observed. Respiratory compensation point represents the point at 1) which $\dot{V}_E$ increases disproportionally to $\dot{V}CO_2$, 2) the second deviation of the linear increase in $\dot{V}_E$ and 3) which an increase in both $\dot{V}_E/\dot{V}O_2$ and $\dot{V}_E/\dot{V}CO_2$ is seen (21). These breakpoints were corrected for the $\dot{V}O_2$ mean response time (22).

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3-km TT. Split times were taken at each 250-m interval. Completion time per 250 m was calculated by subtracting the split time at the previous time point from the time at the respective 250-m point.

Normalized power was determined by taking the value of the 30-s rolling average at each point to the fourth power, averaging this and taking the fourth root of this average (23). Power per 250-m bout was defined as the average power over the 250-m interval.

Constant load exercise trials. For $\dot{V}O_2$, breath by breath values were first edited by removing aberrant data (lying 3 SD from the local mean). Ninety-five percent confidence bands were used to identify these points. Second, single trials were linearly interpolated on a second-by-second basis and data coming from different trials were mediated to reduce signal noise during transition after exercise onset (24). Third, these data were time averaged as 5-s bins. For NIRS, second by second data were used.

Data for $\dot{V}O_2$ and [HHb] kinetics were analyzed similar to Murias et al. (25) using Origin 2019b (OriginLab, Northampton, MA). A monoexponential function (1) was used to model on-transient responses for both $\dot{V}O_2$ and [HHb]:

$$Y_{(t)} = Y_{BL} + A_p\left(1 - e^{-(t-TD)/\tau}\right) \qquad [1]$$

with $Y_{(t)}$ the value of the dependent variable at any particular time during the transition. Y_{BL} stands for the baseline value before transition, whereas A_p represents the amplitude of the increase in Y. τ is the time constant of the response, which is the time needed to increase Y to 63% of the absolute increase in Y, whereas TD stands for the time delay. For $\dot{V}O_2$, the cardiodynamic phase (phase I) was removed from the fitting on a visual basis. Mean response time (MRT) for [HHb] and $\dot{V}O_2$ was calculated as the sum of TD and τ.

Statistical Analysis

All data were expressed as mean ± SD. Subjects served as their own controls. IBM SPSS statistics 24 package was used for statistical analysis. Shapiro–Wilk test was used to control for normal distribution of the data, while Mauchly's test of sphericity indicated that sphericity was not violated. Statistical significance was set at $P < 0.05$ for all statistical tests.

Acute Hb response. A one-way repeated measures (RM) MANOVA was used to evaluate the general acute Hb response for all 12 apnea protocols.

A 2 × 3 × 2 repeated measures MANOVA (mode × frequency × intensity) for the measures Δ0 and Δ5 was used to determine differences in the acute Hb response for all three modalities. Pairwise comparisons, using the Bonferroni test, were applied as *post hoc* to compare the differences between the independent within-factors. Sphericity was not violated for any of the variables so no corrections were applied.

Psychological response. Perceived readiness rating and BORG for static maximal conditions (S1x, S5x, and S5x + 1) were analyzed using Wilcoxon tests.

Warming up protocol. A 2 × 3 RM ANOVA (condition × time point) was used to determine the influence of warming-up on [Hb] and whether this response differed between conditions. For significant effects, pairwise and 2 by 2 comparisons were used to look deeper into where the differences were situated.

3-km TT. A one-way RM MANOVA was used to determine differences for the three conditions (control [CO], placebo [PB], apnea [AP]) in the variables completion time, average power and speed for each 250-m bout and performance time, average cadence and power, peak HR, peak $\dot{V}O_2$, and end lactate.

Constant load cycling. A one-way RM MANOVA was used to investigate differences between conditions (apnea, control) in kinetic parameters for pulmonary oxygen uptake (baseline $\dot{V}O_2$, phase 1, phase 2, amplitude, time delay [TD], τ, and MRT) and peripheral muscle extraction (baseline [HHb], amplitude, TD, τ, and MRT).

RESULTS

Part 1: In Search of the Optimal Apnea Modalities

General Hb response. An overview of the general response to static apnea is presented in Figure 2A. An immediate increase in [Hb] after apnea compared with baseline was seen in all protocols ($P < 0.05$). Five minutes after the last apnea, [Hb] was still significantly higher in all static protocols ($P < 0.05$) except for the S5x 80% protocol ($F = 0.985$, $P = 0.148$).

Data on the response to dynamic apnea are presented in Figure 2B. Fifteen minutes of cycling elevated baseline [Hb] (BL) from 147.5 ± 9.1 $g \cdot L^{-1}$ to 150.9 ± 7.1 $g \cdot L^{-1}$ after cycling ($F = 106.358$, $P < 0.001$). An immediate increase in [Hb] after the last apnea was seen in all dynamic protocols ($P < 0.05$), except for the D5 + 80% protocol ($F = 0.542$, $P = 0.297$). Five minutes after the last apnea, [Hb] was no longer significantly higher ($P > 0.05$) in any condition. Moreover, [Hb] in D5 + 80% ($F = 0.578$, $P = 0.003$) and D1x80% ($F = 0.593$, $P = 0.005$) was significantly lower 5 min postapnea as compared with preapnea.

Main effects. The 2 × 3 × 2 RM MANOVA revealed main effects for Mode and Intensity at both Δ0 and Δ5. For Frequency, an effect was only found at Δ5. No significant interaction effects were observed ($P > 0.05$).

For Mode, static conditions evoked stronger responses than dynamic conditions both immediately after (Δ0: $F = 9.093$, $P = 0.015$) as well as 5 min postapnea (Δ5: $F = 28.349$, $P < 0.001$).

For Intensity, maximal apneas were better at evoking a strong Hb response than submaximal (80%) at both Δ0 ($F = 34.439$, $P < 0.001$) and Δ5 ($F = 9.321$, $P = 0.014$).

For Frequency, no significant difference was found between the conditions at Δ0 ($F = 2.615$, $P = 0.101$) indicating that the magnitude of the Hb response is similar for one, a series of five and five plus one apnea. However, for Δ5, a main effect was apparent ($F = 15.512$, $P < 0.001$). Pairwise comparisons indicated that the series of five apneas was significantly better than a single apnea ($P = 0.018$), whereas the five plus one condition

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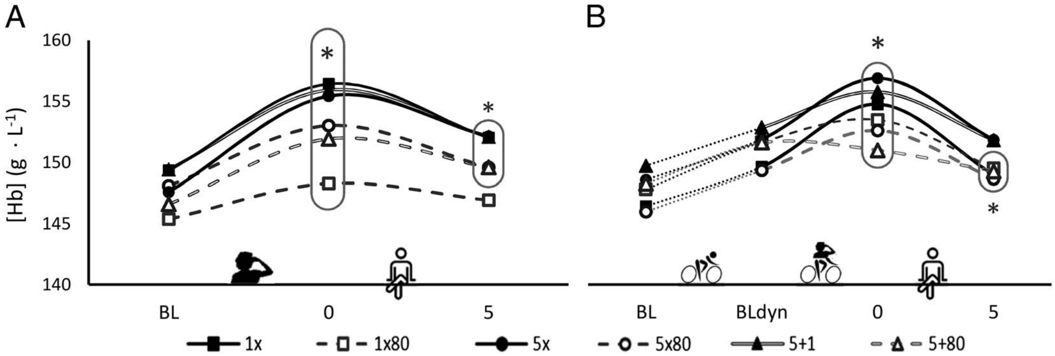

FIGURE 2—Acute Hb responses for both static (panel A, *n* = 11) and dynamic protocols (panel B, *n* = 10). ***Solid lines*** **represent maximal apneas, whereas** ***dotted lines*** **exemplify 80% apneas. BL, baseline; BL_{dyn}, baseline after 10 min of cycling; 0, immediately after the apneas; 5, after 5 min of seated rest following apnea. *Significantly different (*P* < 0.05) from BL (panel A) and BL_{dyn} (panel B) for all encircled values.**

was better than one single apnea (P = 0.002) and a series of five + one apnea did not differ from a series of five (P = 0.108).

Perceived exertion and readiness for static conditions. Based on the physiological values, static maximal apneas are needed to evoke the best response, both immediately after as well as 5 min after apnea. Although there is no significant difference immediately postapnea for the different amount of apneas, the magnitude of the Hb response remains larger after 5 min when five apneas were performed. On the psychological side, subjects indicated to be readier to perform in a competition both immediately after as well as 5 min after apnea after one apnea compared with a series of five, whereas no differences were seen with the S5x +1 condition. Combining both physiological and psychological factors, therefore, suggests that performing one single maximal apnea shortly before exercise shows the best promise to improve performance.

Part 2: The Influence of Apnea on Performance

Warming-up protocol. An interaction effect revealed (F = 3.475, P = 0.049) differences between apnea and control for the Hb pattern (Fig. 3). After an increase due to warming-up, [Hb] decreased in both conditions (apnea and control), but remained elevated. For the apnea condition, [Hb] increased from 152.1 ± 8.9 to 162.1 ± 9.1 $g{\cdot}L^{-1}$ after warming-up and decreased to 158.9 $g{\cdot}L^{-1}$ (P < 0.001) after 10 min of rest with implementation of apnea. In the control condition, an increase from 145.5 ± 10.3 $g{\cdot}L^{-1}$ to 157.4 ± 9.2 $g{\cdot}L^{-1}$ and decrease to 152.4 ± 10.1 $g{\cdot}L^{-1}$ (P < 0.001) was seen. The decrease in [Hb] was attenuated after apnea (P = 0.038, F = 5.569) indicating a small, but significant apnea-specific effect.

3 km TT. With completion times of 263.9 ± 12.9 s for control, 264.0 ± 15.8 s for placebo, and 264.8 ± 14.1 s for the apnea condition, no differences in 3-km TT performance could be observed (F = 0.250, P = 0.679). Logically, average normalized power output (control, 329 ± 43 W; placebo, 332 ± 48 W; apnea, 327 ± 44 W, F = 1.711, P = 0.270) and average cadence (F = 1.287, P = 0.298) did not differ either. Furthermore, no differences were found for peak HR (control, 193 ± 7 bpm; placebo, 194 ± 7 bpm; apnea, 193 ± 8 bpm; F = 0.517; P = 0.575), peak $\dot{V}O_2$ (control, 3667 ± 4 25 $mL{\cdot}min^{-1}$; placebo, 3713 ± 500 $mL{\cdot}min^{-1}$; apnea, 3651 ± 404 $mL{\cdot}min^{-1}$; F = 0.521; P = 0.602), and end lactate (control, 15.9 ± 3.6 $mmol{\cdot}L^{-1}$; placebo, 15.9 ± 2.4 $mmol{\cdot}L^{-1}$; and apnea, 14.9 ± 3.3 $mmol{\cdot}L^{-1}$; F = 1.400 and P = 0.270).

Figure 4 gives an overview of the completion times and average power for each 250-m bout. No significant differences between conditions were found in the first 2750 m (P > 0.05), whereas subjects pedaled at a lower power output in the last 250 m (F = 5.940, P = 0.01) in the apnea condition compared with control (F = 12.459, P = 0.003) and PB (F = 19.063, P = 0.065). This is reflected in the faster completion time in the last 250 m in PB (21.1 ± 1.2 s) and control (21.1 ± 1.5 s) as compared with apnea (22.1 ± 2.0 s, F = 3.590, P = 0.049).

Constant load trials. The pattern of [HHb] extraction and oxygen uptake can be seen in Figure 5. A trend toward faster $\dot{V}O_2$ on-kinetics in the apnea protocol was seen for τ (F = 5.029, P = 0.06) and MRT (F = 4.807, P = 0.064), whereas TD was similar for both conditions (F = 0.021, P = 0.667). TD was significantly shorter for [HHb] (F = 8.647, P = 0.022), τ

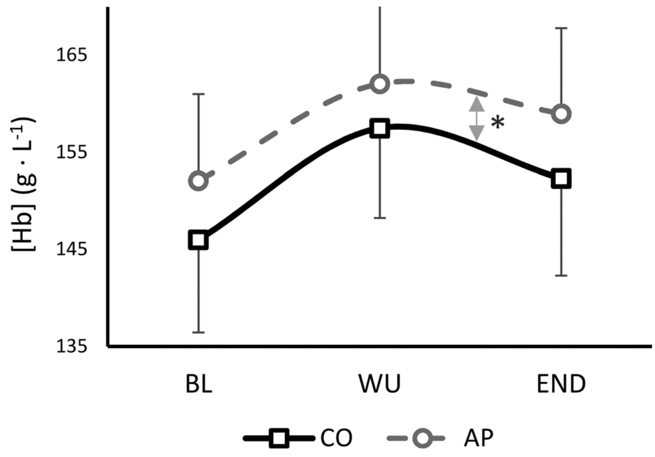

FIGURE 3—Mean (*n* = 12) and SD for [Hb] during warming-up protocol at baseline (BL), at the end of the 10-min warming-up (WU) and after 10 min of rest (END) for the control condition (CO, ***solid line*****) during which subjects remained seated and the apnea condition (AP,** ***dotted line*****) during which subjects performed one maximal static apnea. *Significantly different from baseline (*P* < 0.05).**

APPLIED SCIENCES

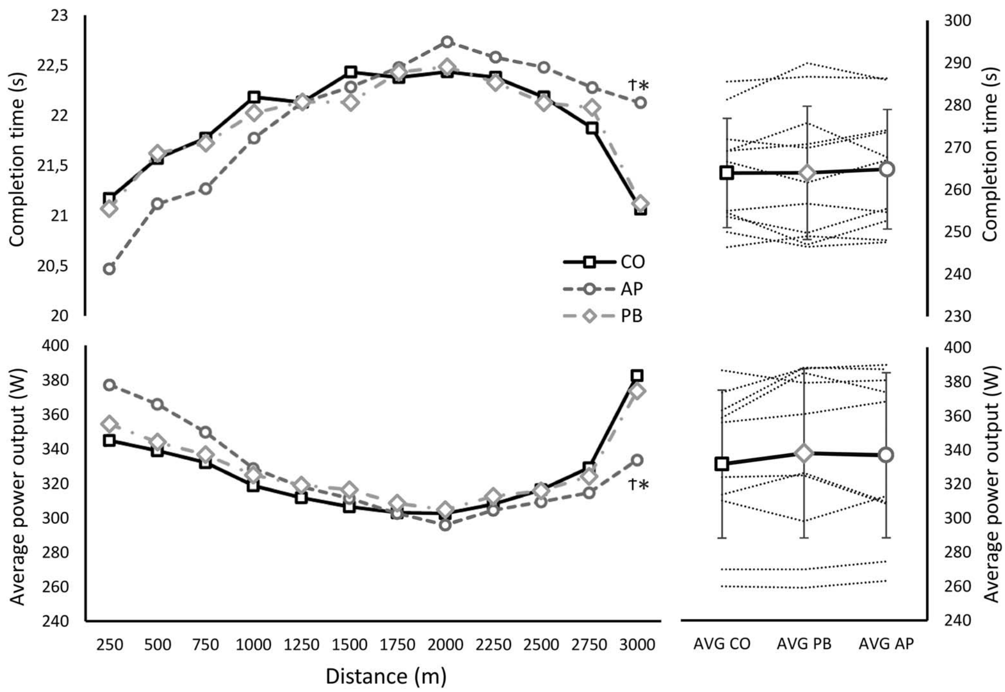

FIGURE 4—Mean (n = 10) values for average completion time (upper graph) and power (lower graph) for each 250-m bout and the average (n = 11; group average, *solid black line*; individual values, *dotted lines*) of the entire trial for all three conditions: CO, PB, AP. *Significantly different from CO (P < 0.05). †Significantly different from PB (P < 0.05).

and MRT did not differ (F = 2.156, P = 0.185, F = 4.470, P = 0.072, Fig. 5).

DISCUSSION

This study thoroughly investigated the potential of apnea to boost immediate performance. First, apnea modalities with the highest increase in [Hb] and lowest negative impact on perceived readiness were selected. Our results suggest that one single maximal apnea shortly before performance shows the highest potential to improve performance. Second, despite the observation that performing only one maximal static apnea tended to speed $\dot{V}O_2$ kinetics in work performed at the heavy intensity domain, no effect on a 3-km TT performance was seen when performing a static apnea integrated in a warming-up.

Part 1: In search of the optimal modalities. The present study is the first to compare the influence of different apnea modalities on the Hb response. This study shows that maximal

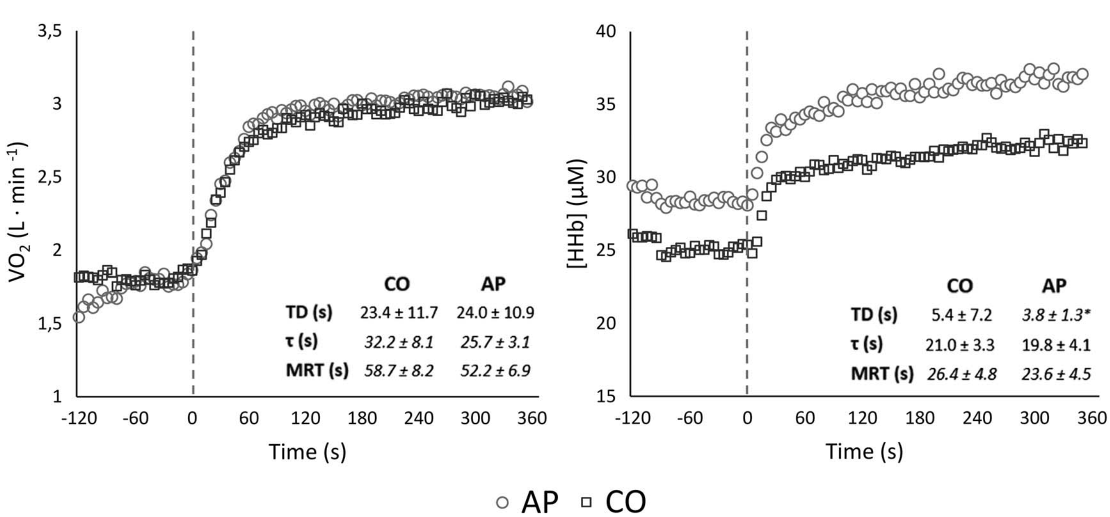

FIGURE 5—Mean (n = 8) values for oxygen uptake ($\dot{V}O_2$) and extraction (Δ[HHb]) during the constant work tests at P = Δ25. *Circles* represent the AP condition, *squares* visualize the CO condition. The small table presents the mean (n = 8) and SD for $\dot{V}O_2$ and [HHb] on-kinetics parameters for CO and AP during a 6-min constant load at a work rate of P = Δ25. A_p, amplitude of the response; τ, tau or time constant. *Significantly different from CO (P < 0.05). Italics indicate P < 0.1.

apneas evoke a stronger increase in [Hb] than submaximal apneas. This does not only indicate that the duration/intensity is an important factor for the spleen response but also suggests that the spleen contraction and corresponding boost in [Hb] develop until the end of apnea. This is in contrast to the findings of Bakovic et al. (26) who observed an immediate contraction at onset of apnea followed by a more steady leveling off. Based on Bakovic's findings, the magnitude of the [Hb] response for protocols at 80% of maximal breath hold time would be expected to be similar to maximal apneas, while our results indicate the opposite. However, a similar immediate strong decrease in spleen volume followed by a more steady decrease was also seen in elephant seal pups (27), whereas hematocrit increased almost linearly to dive time. Thornton et al. (27) argued that, although the spleen had already released most of its RBC after 2 min, the volume of the hepatic sinus increases which delays the release of RBC into the blood circulation in elephant seals. Although this has not yet been confirmed in humans, this mechanism could explain the contradiction between our findings and the spleen response observed by Bakovic et al. (26). This leads us to believe that a longer duration is necessary to enable both the spleen and hepatic sinus to optimally release RBC into the blood circulation. Additionally, this could also explain why dynamic apneas appear less efficient to increase [Hb] than static apneas. During dynamic apnea, average maximal apnea duration was 45 s, while for static apnea the maximal apnea duration was on average 147 s, which means that not only the spleen, but also the hepatic sinus has up to three times more time to release RBC.

Based on the present results, it is clear that maximal static apneas show the highest potential to improve performance. Results are, however, less clear for the modality frequency as there is no difference in effect for one or a series of five apneas. In this context, Bouten et al. (3) showed a steep first decrease in spleen volume after the first apnea followed by a more graduate leveling off after the fourth of a series of five apneas, indicating that one apnea is sufficient to induce most of the blood-boosting spleen contraction. Although in the present study, no difference in Hb response was found immediately postapnea for the different frequencies, the response appeared to last longer after a series of five apneas. It could, therefore, be argued that a series of five maximal static apneas performed as close as possible before exercise should be used when testing for the effect on performance. Indeed, the physiological response appears to last longer for this particular protocol albeit the advantage compared with only one apnea was not significant. Moreover, psychological factors are also important, which is supported by the results of Sperlich et al. (16). Accordingly, subjects gave the five-apnea protocol a higher rating of perceived exertion and indicated to be less prepared for exercise performance in comparison to only one apnea. A holistic approach, therefore, indicates that one single maximal static apnea would be most suited to implement in a warming-up routine before competition. This is especially true when considering not only physiological but also psychological and practical factors.

Part 2: The influence of apnea on performance. One single maximal static apnea tended to be effective in speeding $\dot{V}O_2$ kinetics (i.e., τ and MT) by 6.5 s (20% for τ and 11% for MRT) compared with the control condition at a power output corresponding to Δ25. This observed effect is similar to other interventions that have increased the O_2 supply to the muscle, such as MacDonald et al. (14) who found a 25% (6.0 s) speeding of τ after hyperoxic breathing as compared with normoxic breathing. Both increases could be ascribed to a better transport of O_2, either through more O_2 solved in the blood or through a higher [Hb] and thus higher oxygen binding capacity, and therefore better delivery to the working muscle. Indeed, results from part 1 of this study show that one single maximal apnea increases [Hb] by 4% which corresponds with an increase of 7 $g \cdot L^{-1}$. This is in contrast to Wilkerson et al. (28) who did not find faster $\dot{V}O_2$ kinetics after recombinant human EPO treatment increasing [Hb] by 10 $g \cdot L^{-1}$. Whether oxygen delivery is a limiting factor in $\dot{V}O_2$ kinetics in healthy, physically active males is still in debate. According to the Tipping Point model (24), there is a critical "tipping point" at which $\dot{V}O_2$ kinetics are no longer limited by O_2 delivery and, therefore, improved delivery does not speed kinetics. This is supported by several studies showing no effect after recombinant human EPO administration (28) and hyperoxia (29), although our results and data from more recent hyperoxic studies did show improvements, more specifically above GET (1,14). In the present study, it was also found that muscle [HHb] was elevated at the onset of the constant load exercise (CLE) in combination with faster [HHb] kinetics (i.e., 30% shorter TD) in the apnea condition. This is likely the result of the apnea-induced bradycardia and peripheral vasoconstriction, limiting O_2 supply to the muscles and as such increasing fractional O_2 extraction (10). This shows that these apnea-related physiological responses induced muscle tissue deoxygenation, which might not have been fully reversed at the onset of exercise. It is possible that these faster [HHb] kinetics, reflecting faster O_2 extraction, contribute to the faster $\dot{V}O_2$ kinetics observed during the CLE.

Despite the effectiveness of one single maximal static apnea in increasing [Hb] by 4.5% and $\dot{V}O_2$ kinetics by 20%, the absence of an effect on the 3-km TT when integrated in a warming-up is very straightforward. There was no difference in time to complete the TT between the control, placebo, and apnea condition. Accordingly, neither average power output nor average cadence was shown to be different. Only split times might indicate a different pacing: subjects tended to start faster in the apnea condition but could not keep up until the end. However, interindividual variation is large, and therefore, statistical evidence to support this claim is very limited. This is in agreement with Sperlich et al. (16) who did not find differences after a series of four apneas in 4-km TT performance nor split times. Robertson et al. (17) also did not find a difference between apnea and control conditions for a 400-m swim race. Additionally, the underpinning physiological mechanisms do not show differences either: peak HR, peak oxygen uptake and end lactate were similar during all trials, explaining the lack of effect of apnea on performance. This is

APPLIED SCIENCES

in line with Sperlich et al. (16) who did not find differences for lactate, HR, and other physiological parameters.

There are several reasons why physiological factors and performance had not improved, whereas [Hb] showed a slight, but positive pattern and $\dot{V}O_2$ kinetics tended to be faster. First, the time window during which the increase in [Hb] creates an advantage might be too narrow. At the onset of the 3-km TT, the spleen will also contract in response to intense exercise (10). The advantage of starting with increased [Hb] through apnea seems not sufficient to improve performance. Second, the potential benefits could be offset by the warming-up. Although one static maximal apnea increased [Hb] from 149.3 to 156.0 $g \cdot L^{-1}$, the magnitude of the response appears much smaller when implemented in the warming-up and might, therefore, be too small to influence performance as the warming-up itself also evokes a spleen contraction which did indeed lead to an increase in [Hb]. However, in contrast to apnea, the increase during warming up is a combination of spleen contraction and hemoconcentration: an exercise induced decrease in plasma volume artificially increases [Hb] without an actual increase in Hbmass. The increase after apnea, on the other hand, can be almost entirely ascribed to spleen contraction and, therefore, leads to an actual increase in Hb (30). Additionally, we expected both effects (hemoconcentration and spleen contraction) to be mostly reversed after 10 min. Data from the control condition indicate otherwise, potentially explaining why the response in [Hb] to acute apnea was less pronounced when integrated in the warming-up. More physiologically challenging apnea protocols might, therefore, be needed; however, the psychological and practical disadvantages advocate for a more simple approach. Additionally, a warming-up already speeds $\dot{V}O_2$ kinetics and has been proven to improve 3-km TT performance (31) as well as other types of performance (32,33). Therefore, this priming effect may overrule the possible gain through apnea. Comparatively, Robertson et al. (17) found that three maximal static apneas integrated in a warming-up improved 400-m swimming performance significantly compared with performance without both warming-up and apnea. However, only a minor and insignificant benefit was seen compared with the warming-up protocol without apnea. Third, it has to be considered of course that $\dot{V}O_2$ kinetics were analyzed in the heavy-intensity domain, whereas 3-km TT performance is situated in the severe domain. Kinetics were not analyzed for 3-km TT because the pacing profile showed no constant power output. However, both visual inspection and analysis of $\dot{V}O_2$ at different time points do not indicate differences in the $\dot{V}O_2$ response during the 3-km TT. Additionally, our study included a homogeneous population of subjects who were well trained and had an average $\dot{V}O_2$ peak of 52.5 $mL \cdot min^{-1} \cdot kg^{-1}$. This is a limitation of the study indicating that conclusions cannot automatically be transferred to either untrained individuals or elite athletes with high $\dot{V}O_2$ max in whom physiological systems, such as oxygen transport, delivery, and utilization, are already highly optimized as they might respond differently to apnea.

GENERAL CONCLUSION

Although acute apnea showed promise as a quick and easy to implement method to improve immediate performance, the reality is more complex. Because of the integration of multiple physiological mechanisms occurring simultaneously, this method does not provide an added benefit to performance when added to a warming up, at least for short maximal efforts in a well-trained population.

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